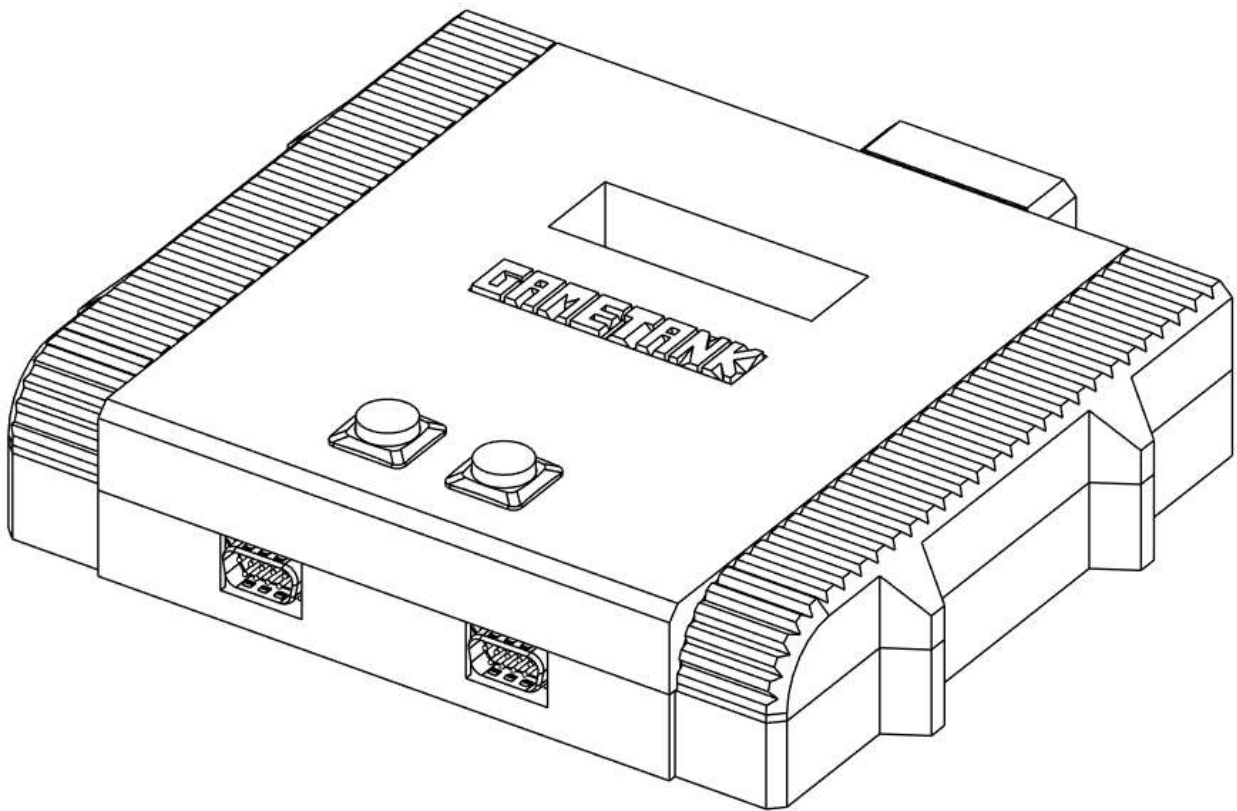


GAMETANK

Manufacturing Brief



Prepared by Clydeware, LLC
Updated 2nd July 2026

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Hardware Descriptions

Main Game Console Unit

A GameTank console consists of two PCBs inside a plastic enclosure. The two PCBs are connected electrically by a pair of flat flexible cables, and mechanically by four standoff fasteners. There is also a 3D-printed spacer placed in the center of the board stack. The enclosure consists of four printed parts, and two push-buttons. The buttons have a wire harness soldered on, which plugs into the motherboard in three places. The bottom half of the enclosure is one solid piece. The upper half of the enclosure has a printed decorative “badge” piece which is thermally bonded to the main upper part of the enclosure body. Lastly, a “vent” cover is placed at the rear of the console. The enclosure is held together by four screws which self-tap into the plastic.

Controller

The input device consists of an assembled PCB, a cable with an attached molded flangeless D-shape connector, four 3D printed parts, two screws, and two spacers. The printed parts include the main body of the controller in which the PCB rests, a faceplate with holes for buttons, a flexure holding four buttons and a plus-shaped directional pad, and an insert to be placed beneath the directional pad that prevents simultaneous opposite direction inputs. The cable is soldered directly to the PCB, and has a rubber strain relief boot that mates with a tab inside the controller body. The two screws are installed from the back of the controller through all other parts, and into the faceplate. The spacers support the center of the flexure, above the PCB. The faceplate is designed so that the screws self-tap.

Data Cartridge

The data cartridge consists of an assembled PCB, a 3D printed shell in two parts, and two screws. The PCB has plated finger contacts for use with the edge connector in the Console and the Flasher. The side of the PCB with mounted components is considered the front of the cartridge. The two halves of the shell have diagonal tabs to allow the cartridge to be stably held together by the two screws near the top. The screws should fit loosely through the back half of the shell, and self-tap into the front half. The cartridge may be affixed with a label bearing artwork for a particular game title. The front half of the shell has a gap to allow the label to be folded over for improved long term adhesion.

If being used as distribution media for software, the Cartridge Flasher Tool is used to load data into the flash memory of the cartridge.

Cartridge Flasher Tool

The cartridge flasher tool consists of an assembled PCB, a 3D printed base, and two screws. The AVR microcontroller on the PCB requires a bootloader and firmware. The programming header is exposed both as header pins and as Tag-Connect contacts. If the Tag-Connect header can be used for flashing at scale, the header pins may be omitted.

Bill of Materials

Main Game Console Unit – Motherboard

Qty	MPN	Description
1	5530843-3	Card Edge Connector
2	62684-502100ALF	Connector, Flex Cable
2	74HC04PW,118	Hex inverter
9	74HC257PW,118	Quad 2-to-1 multiplexer
2	74HC40103D,653	8-bit synchronous binary down counter
1	74LVC2G07GW,125	Dual buffer, open drain
1	860020273009	Capacitor 220μF, Electrolytic
1	AS6C4008-55TIN	SRAM, 512K x 8, 55ns
1	AS7C256A-15TIN	SRAM, 32K x 8, 15ns
3	CC0805KRX7R7BB104	Capacitor 0.1μF, 16V, X7R, 0805
1	CD74AC163M96	Binary counter, synchronous, 4-bit
4	CD74HC08PWR	Quad 2-input AND gate
1	CD74HC30PWR	8-input NAND gate
2	CD74HCT238PWR	3-to-8 line decoder/demultiplexer
1	CDBW140-G	Power Diode
2	CL32A106KAULNNE	CAP CER 10UF 25V X5R 1210
2	CL32A226KOJNNNE	CAP CER 22UF 16V X5R 1210
1	DCJ200-10-A-K1-K	DC power jack
3	G800WC306018EU	2-pin header, 2.54mm
1	GRM1555C1H101JA01D	Capacitor 100pF 0402
94	GRM155R62A104KE14J	Capacitor 0.1μF, 100V, X5R, 0402
1	HDL26-SL-B	D-sub connector, 26-pin
2	LD09P13A4GX00LF	D-sub connector, 9-pin female
1	LFSPXO022134	Oscillator / clock generator (Clydeware Provided)
1	LPD5030-103MRC	Common Mode Choke
1	MC74HC1GU04DFT1G	Single Logic Inverter

2	MC74HC32ADTR2G	Quad 2-input OR gate
3	MC74HC74ADTR2G	Dual D-type flip-flop
5	MC74HC86ADTR2G	Quad 2-input XOR gate
1	MCP130T-475I/TT	Voltage supervisor, reset IC, 4.75V
8	MI0805K400R-10	Ferrite Bead
1	RC0805FR-07150RL	Resistor 150Ω, 1%, 0805
16	RC0402JR-1310RL	10 Ohm Resistor 0402
1	RC0603FR-0710RL	10 Ohm Resistor 0603
3	RC0603FR-07100RL	100 Ohm Resistor 0603
9	RC0805FR-073K3L	Resistor 3.3kΩ, 1%, 0805
1	RC0805FR-07150RL	150 Ohm Resistor 0805
11	SN74HC573APWR	Octal latch, transparent, active-high
1	SN74HC139PWR	Dual 2-to-4 line decoder
8	SN74HC163PWR	4-bit binary counter
1	SN74HC640PWR	Octal bus transceiver, inverting
3	SN74HCS00PWR	Quad 2-input NAND gate
1	TPS22917DBVT	Load Switch
1	TPS542025	DC-DC buck converter, 2A
1	TYS252012L220M-10	Inductor 22μH, 2520 size
1	W65C02S6TQG-14	WDC 65C02 CPU, 14MHz, QFP (Clydeware provided)
1	W65C22S6TQG-14	WDC 65C22 VIA, 14MHz, QFP (Clydeware provided)

(Main Game Console Unit- Motherboard, cont'd)

Main Game Console Unit – Audio/Visual Board

2	62684-502100ALF	Flex Cable Connector
1	74HC04PW,118	Hex inverter
2	74HC244PW,118	8-bit Buffer/Line Driver 3-state
2	74HC257PW,118	Quad 2-input multiplexer
1	74HC40103D,653	4-bit synchronous up/down counter
4	74HC4040PW,118	12-stage binary ripple counter
3	74VHC373MTCX	8-bit Latch, 3-state outputs
3	860020273009	Capacitor 220μF, Electrolytic
2	AS7C256A-10TIN	Parallel SRAM 10ns 32k X 8
1	CC0805JPNPO9BN471	470pF Cap 0805
1	CC0805JRNPO9BN101	CAP CER 100PF 50V COG/NPO 0805
1	CC0805JRNPO9BN820	CAP CER 82PF 50V COG/NPO 0805
1	CC0805KRX7R9BB472	4700pF Cap 0805
1	CC0805KRX7R9BB681	0805 680pF Capacitor
1	CC0805KRX7R9BB821	0805 820pF Capacitor
1	CD74HC30PWR	8-input NAND gate
1	CY7C1342-35JCT	Dual ported SRAM 35ns 4k X 8 (Clydeware provides)
3	GRM1555C1H101JA01D	CAP CER 100pF 0402
44	GRM155R62A104KE14J	CAP CER 0.1UF 100V X5R 0402
1	LM358APWR	Dual operational amplifier
2	MC74HC08ADTR2G	Quad 2-input AND gate
2	MC74HC32ADTR2G	Quad 2-input OR gate
3	MC74HC74ADTR2G	Dual D-type flip-flop
1	MC74HC86ADTR2G	Quad 2-input XOR gate
1	MI0805K400R-10	Ferrite Bead
8	RC0402FR-07100RL	0402 100 Ohm Resistor
8	RC0402JR-1310RL	0402 10 Ohm Resistor
1	RC0603FR-07100RL	0603 100 Ohm Resistor
1	RC0805FR-0710KL	0805 10kΩ resistor
1	RC0805FR-07150RL	0805 150Ω resistor
1	RC0805FR-0716KL	0805 16kΩ resistor
14	RC0805FR-071KL	0805 1kΩ resistor, 1%
4	RC0805FR-073K3L	0805 3.3kΩ resistor
1	RC0805FR-07470RL	0805 470Ω resistor
1	RC0805FR-0768KL	0805 68kΩ resistor
1	RC0805FR-0775RL	0805 75Ω resistor
1	RCJ-014	RCA Connector, Yellow (Video Out)
1	RCJ-012	RCA Connector, Red (Audio Out)
1	SN74ACT564PWR	Octal D-type flip-flop with tri-state outputs
2	SN74HC573APWR	Octal D-type transparent latch
1	SN74HC151PWR	8-input multiplexer

2	SN74HC161PWR	Synchronous 4-bit counter
2	SN74HC164PWR	8-bit serial-in parallel-out shift register
5	SN74HCS00PWR	Quad 2-input NAND gate
1	THS7374IPWR	Quad, 2-channel video switch
1	TLC7524CDR	IC DAC 8BIT 16-SOIC
1	W65C02S6TQG-14	65C02 CMOS microprocessor

(Main Game Console Unit- Audio/Visual Board, cont'd)

Main Console Unit – Combined Assembly

Qty	MPN	Description
2	05-50-D-0050-A-4-06-4-T	Ribbon Cable, 0.5mm pitch, 50 pos, 50mm lg, opposite side contacts
1	RP3508ABLKYEYELNS	Pushbutton, Momentary, Yellow (Clydeware provided)
1	RP3508BBLKREDREDNS	Pushbutton, Latching, Red (Clydeware provided)
3	Custom	Red/black wire pair, 2pin XH 2.5mm crimp housing, 25cm
4	R30-3011402	Standoffs, 14mm, M3 thread
4	91292A112	Cap screw, 8mm, M3 thread
4	91292A123	Cap screw, 20mm, M3 thread
4	90591A121	Hex Nut, M3 thread
1	PSAC12R-060	6V DC Power Supply
1	RPA-AB01B-H	US Clip for power supply
1	RPE-AB01B-H	EU Clip for power supply
1	https://www.alibaba.com/product-detail/RCA-TV-Composite-Video-Lead-1_1600507184687.html	RCA Cable pair Yellow/Red plugs

Controller

Qty	MPN	Description
1	SN74HC157DR	Multiplexer IC
8	RC0805FR-073K3L	Resistor
8	KSC2 TE 01J LFS	Tactile Switch
1	https:// www.alibaba.com /product-detail/ New-1-8M-6FT-9- Pin_16002401536 07.html	9-conductor cable with molded D-type flangeless connector
2	91292A115	Cap screw, 16mm, M3 thread

Data Cartridge

Qty	MPN	Description
4	CC0805KRX7R9BB104	Capacitor
1	M29F160FT55N3E2	NOR Flash, Parallel, 16Mbit
1	74HC595D,118	Shift Register
1	74HC244D,653	Non Inverting CMOS Buffer
1	74HC1G04GW,125	NOT Gate
2	91292A112	Cap screw, 8mm, M3 thread

Cartridge Flasher Tool

Qty	MPN	Description
7	CC0805KRX7R9BB104	Capacitor
2	CC0805JRNPO9BN470	Capacitor
1	TAJR475K010RNJ	Capacitor
1	CC0805KRX7R9BB103	Capacitor
1	CC0805JRNPO9BN220	Capacitor
2	LTST-C170KFKT	Orange LED
1	LTST-C170GKT	Green LED
1	MI0805K400R-10	Ferrite
1	USB4110-GF-A	USB-C Port
2	RC0805JR-075K1L	Resistor
3	RC0805JR-07270RL	Resistor
1	RC0805FR-0710KL	Resistor
1	EVP-BFAC1A000	Button
1	5530843-3	Card Edge Connector
1	AVR64DA64-I/PT	Microcontroller
1	FT232RNL-REEL	USB-UART Interface IC

3D Printed Parts

Main Game Console Unit

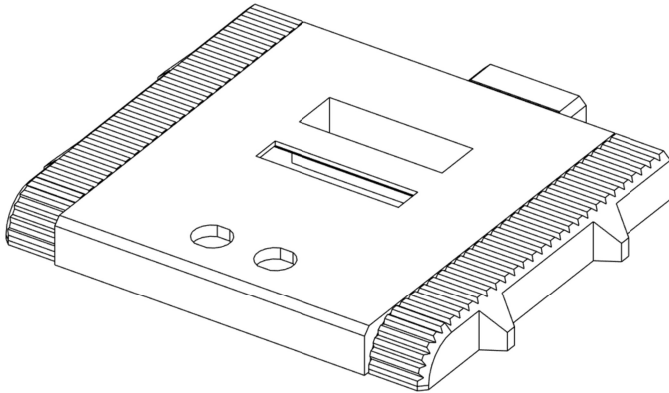


Figure 1: Upper Enclosure Body (Top view)

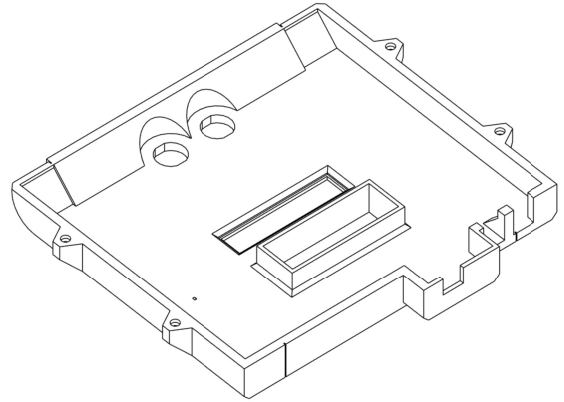


Figure 2: Upper Enclosure Body (Underside view)

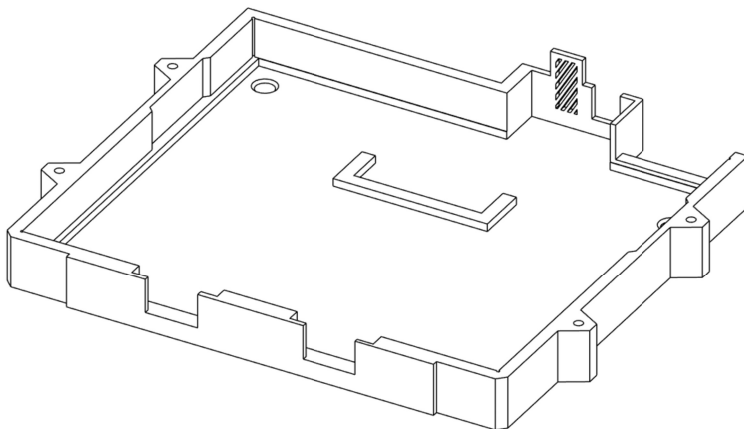


Figure 3: Lower Enclosure Body

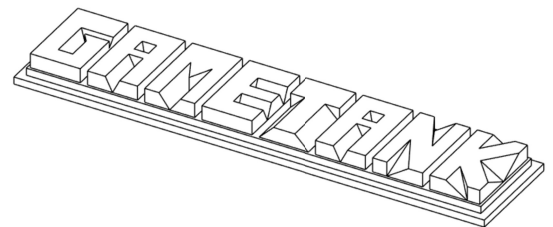


Figure 4: Badge

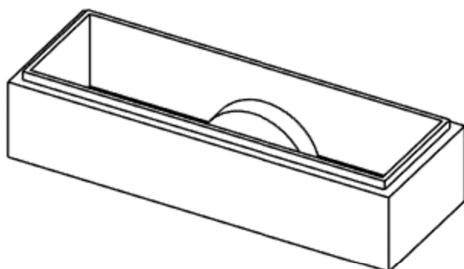


Figure 5: Center Board Support

Controller

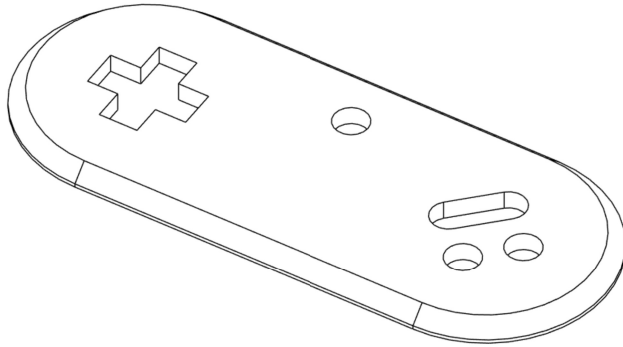


Figure 6: Controller Faceplate

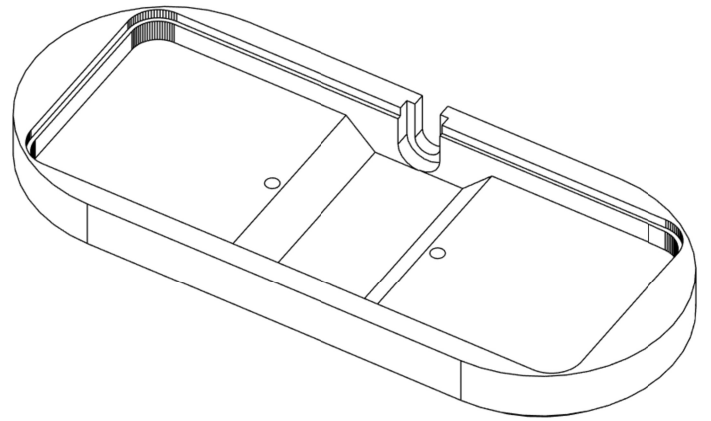


Figure 7: Controller Body

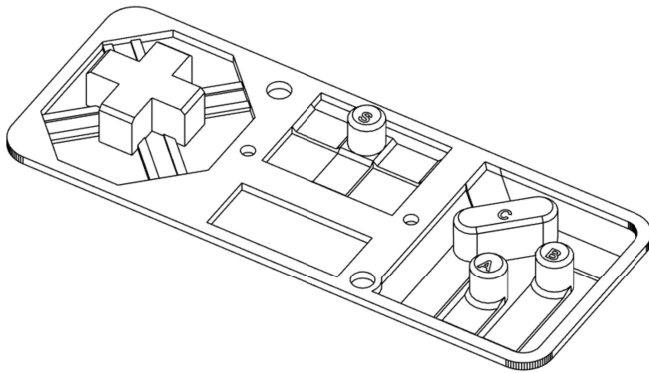


Figure 8: Button Flexure, Multi Material Print Labels

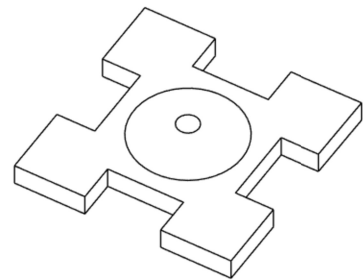


Figure 9: Directional Pad Pivot



Figure 10: Spacers (If printed)

Data Cartridge

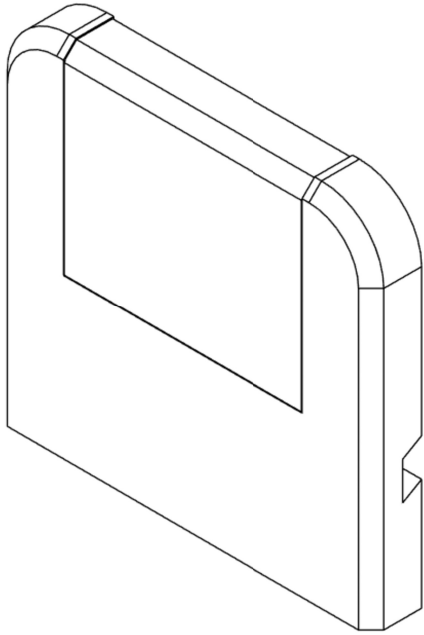


Figure 11: Cartridge Front (Outside)

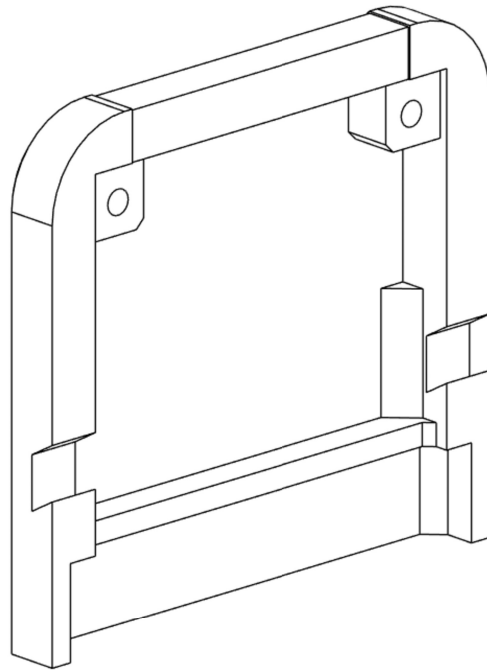


Figure 12: Cartridge Front (Inside)

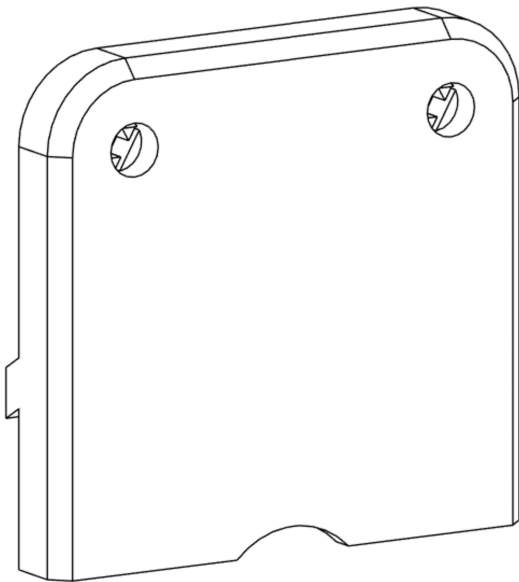


Figure 13: Cartridge Back (Outside)

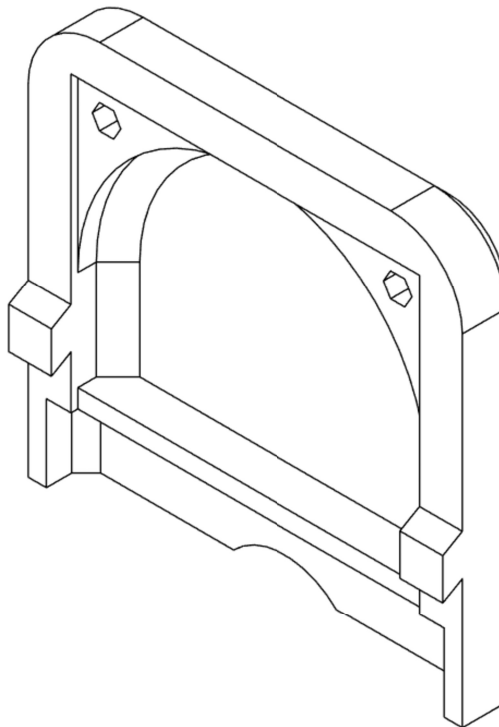


Figure 14: Cartridge Back (Inside)

Cartridge Flasher Tool

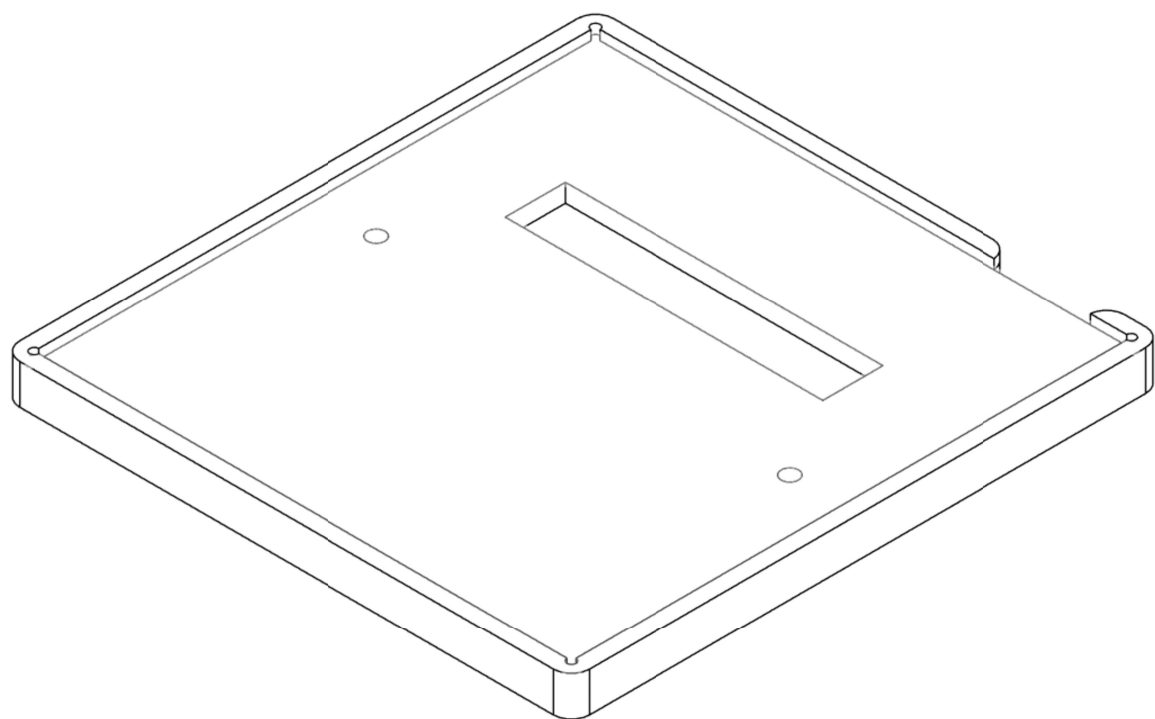


Figure 15: Cartridge Flasher Base

Assembly Procedure

Main Console PCB Stack

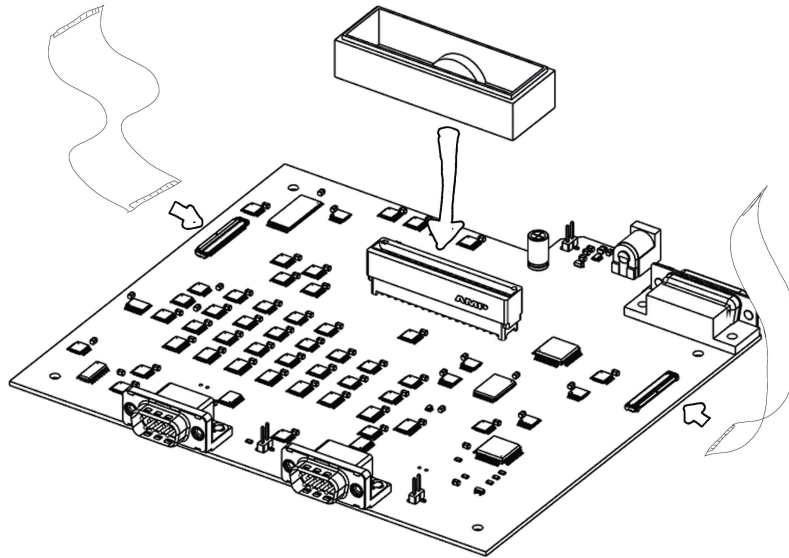


Figure 16: Install FFCs and Cartridge Slot Spacer (Ensure Bump is towards rear)

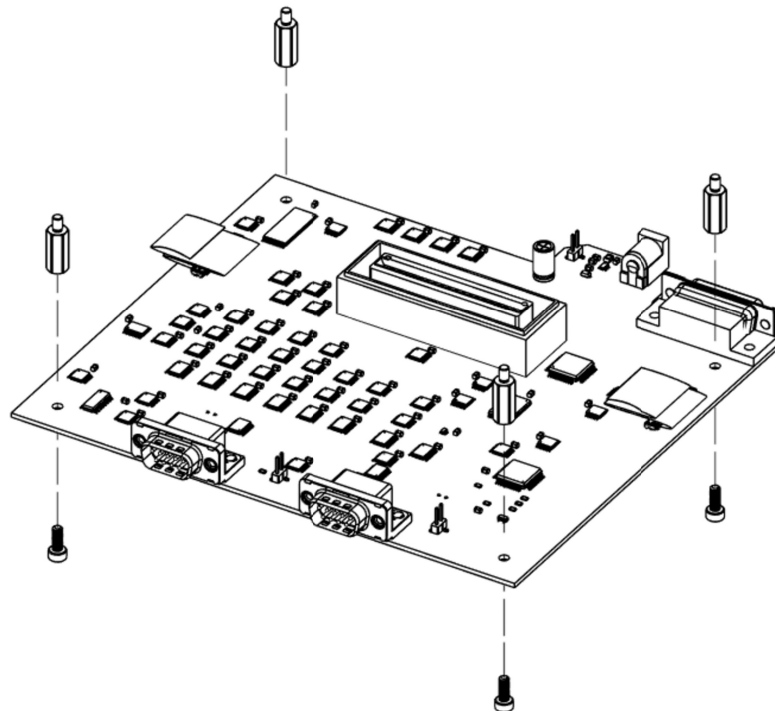


Figure 17: Install standoffs at corners with 8mm screws

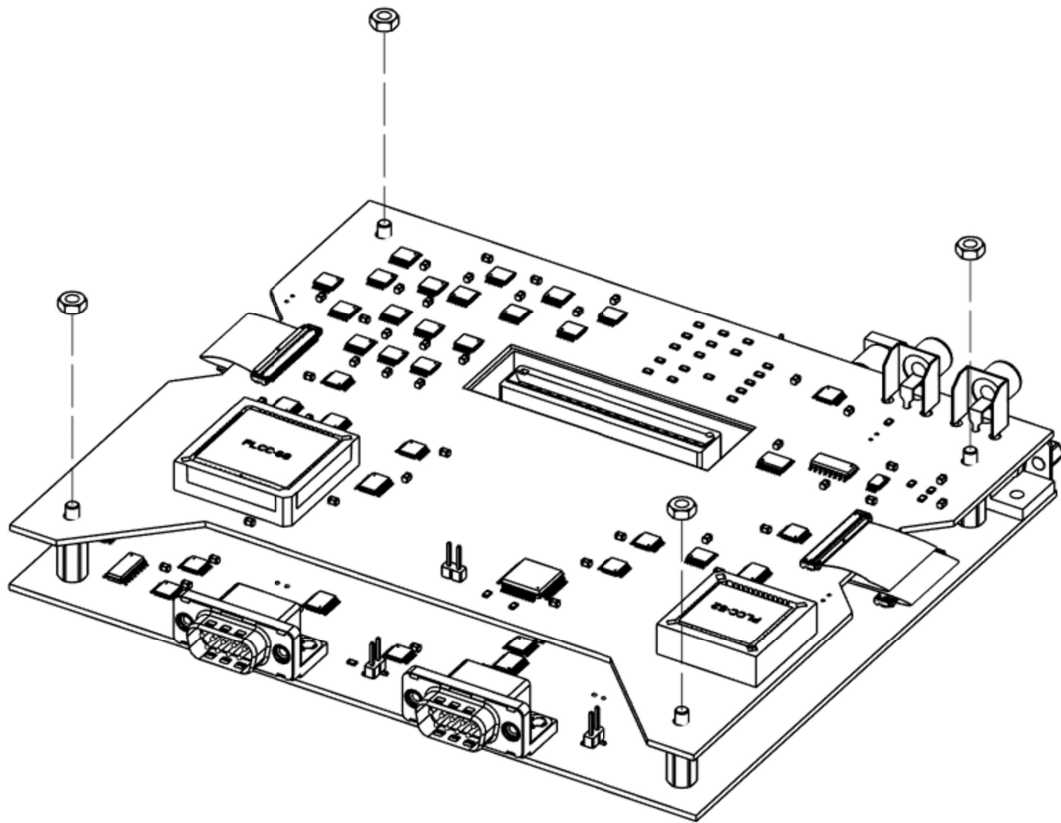


Figure 18: Place video board on standoffs, connect FFCs, and fasten with nuts

NOTE: Illustration features previous version of board, large square chip on left will be absent in current version.

Enclosure Top Half Preparation

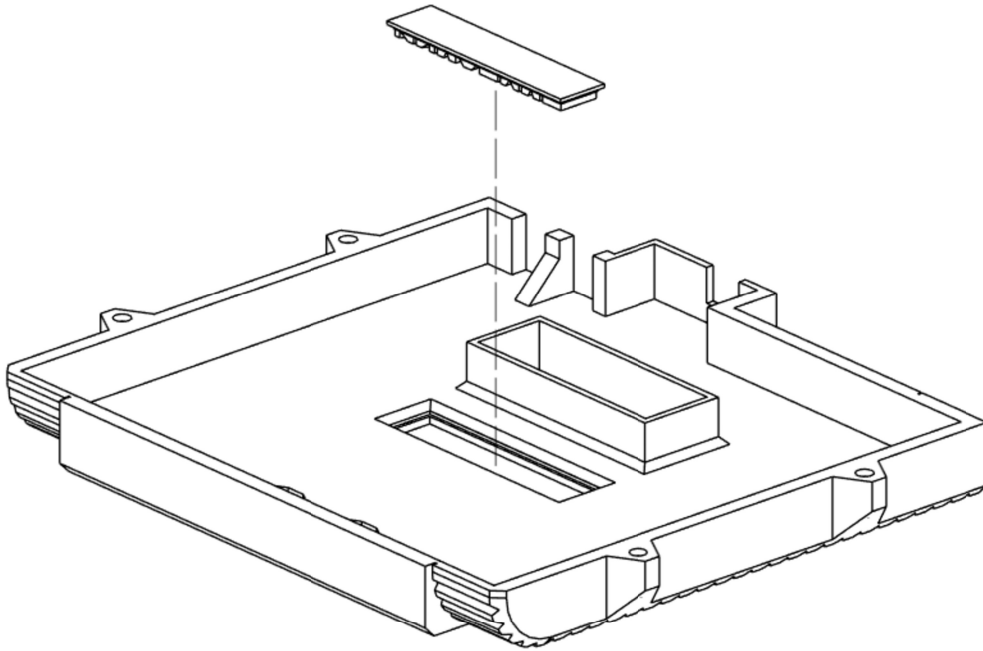


Figure 19: Snap GAMETANK badge into place from inner side

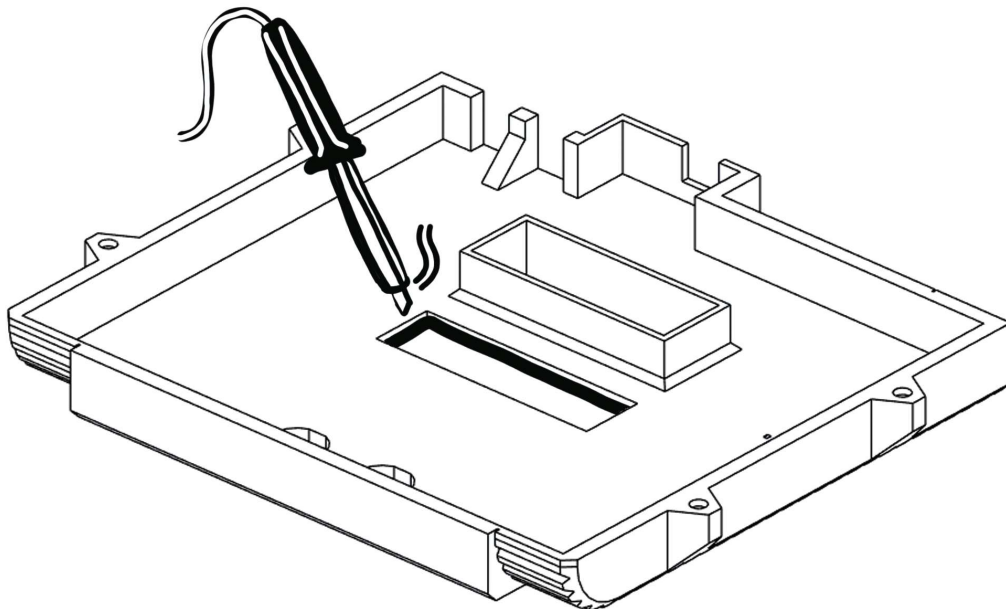


Figure 20: Thermally bond at edges

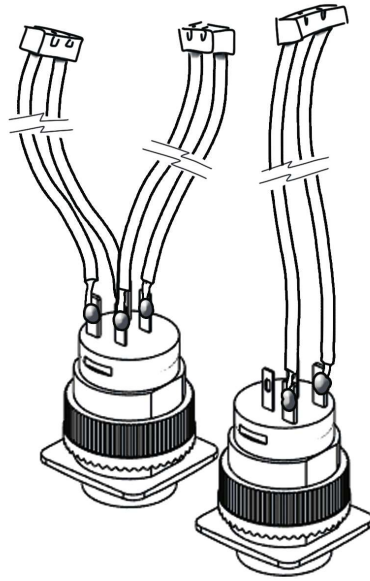


Figure 21: Soldering button wires. Red button gets wires on all terminals. Yellow button gets wires on switched terminals only.

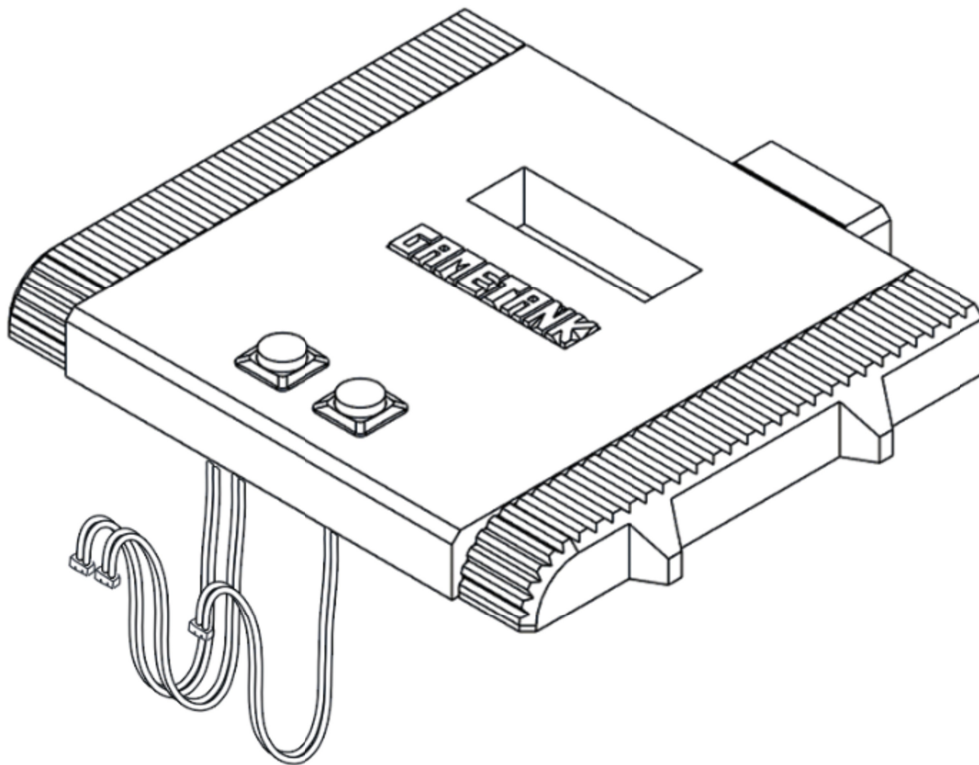


Figure 22: Thread button wires through button holes, fasten buttons with threaded collars

Combining Top Shell, PCB Stack, and Bottom Shell

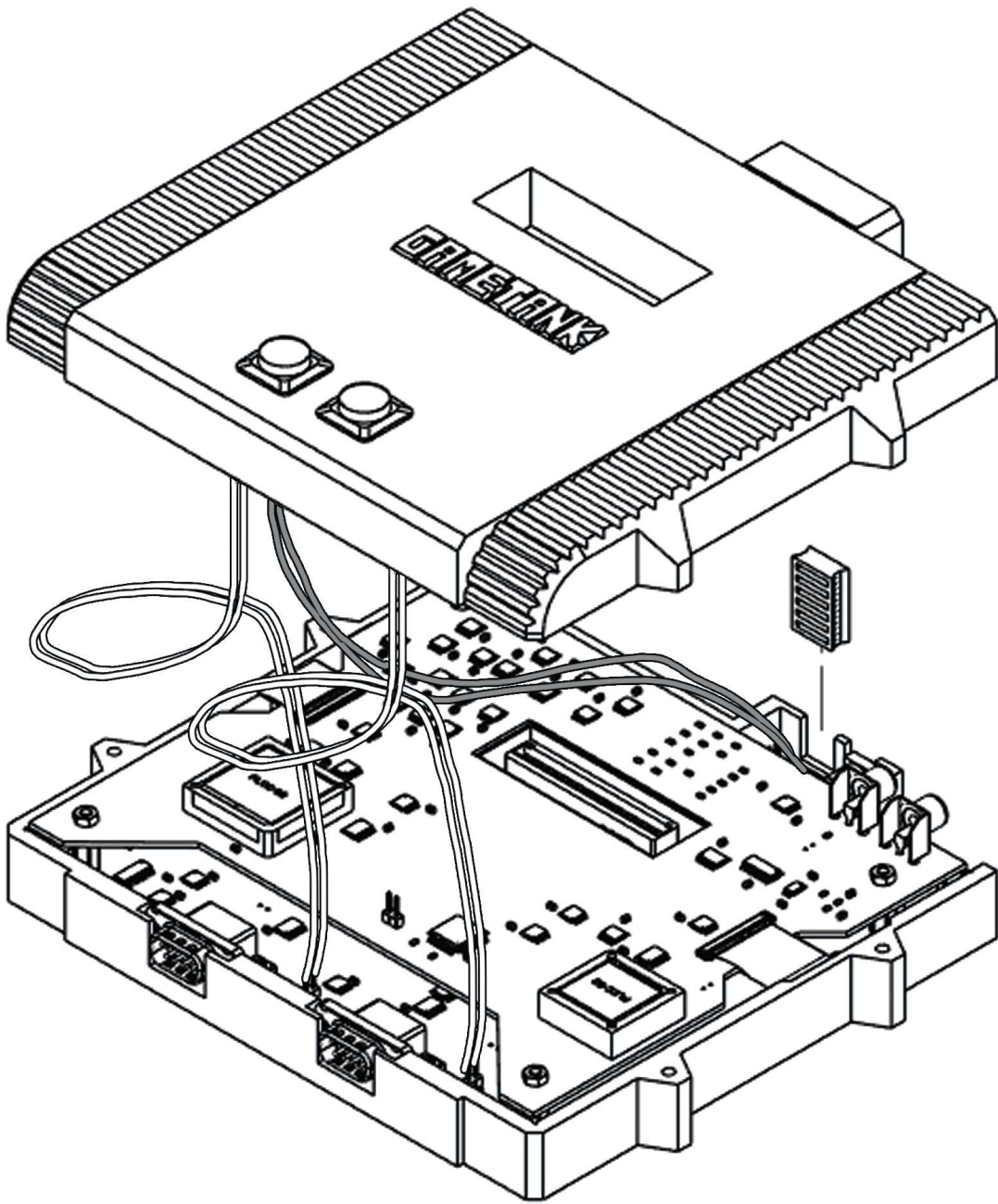


Figure 23: Wires from buttons plug into headers on board. Vent piece is also added.

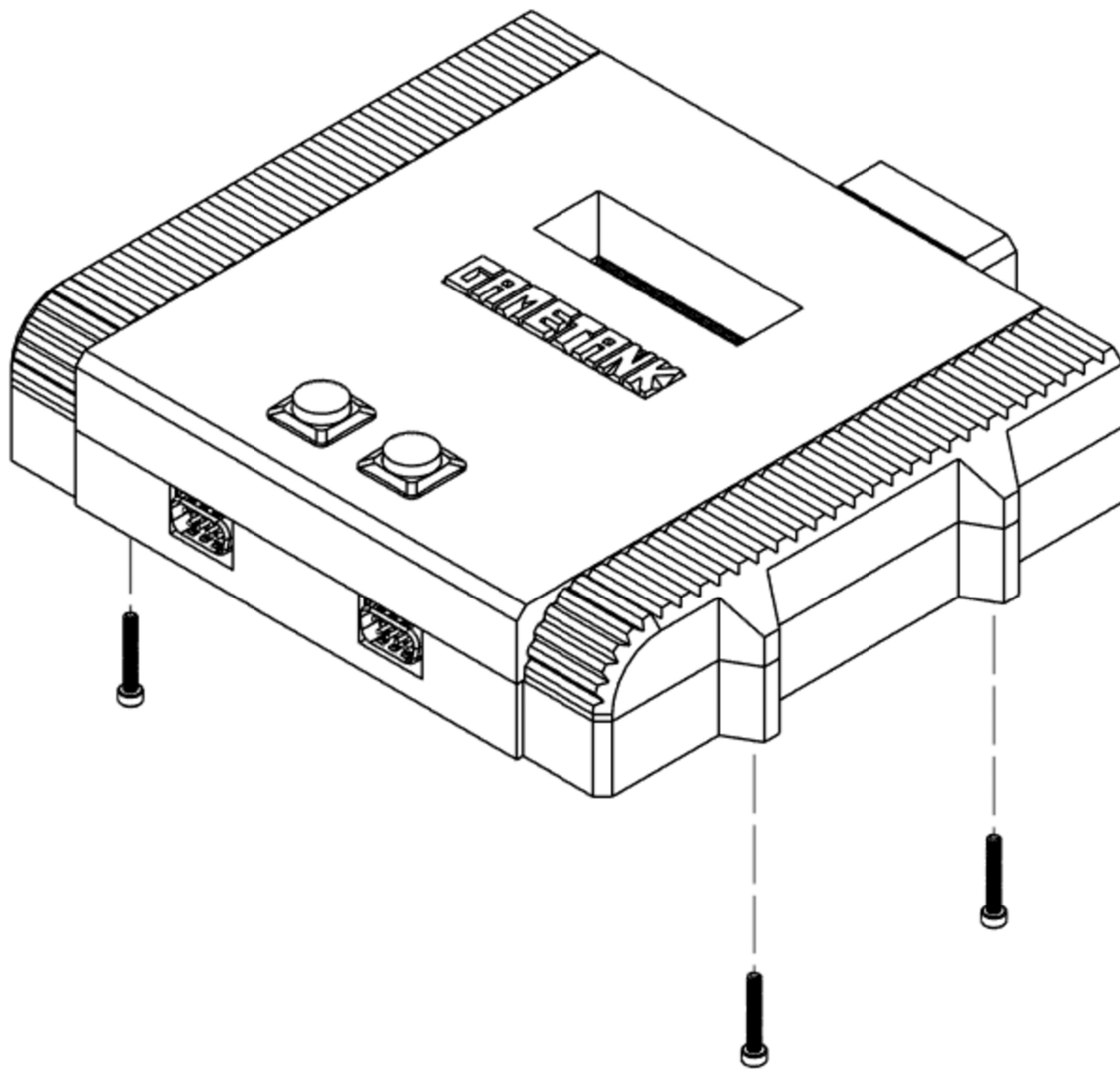
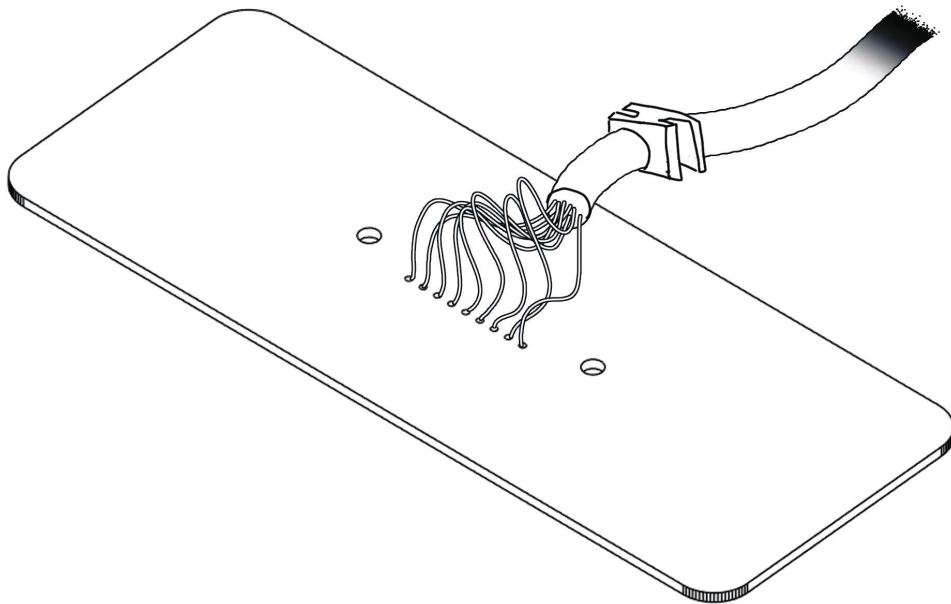


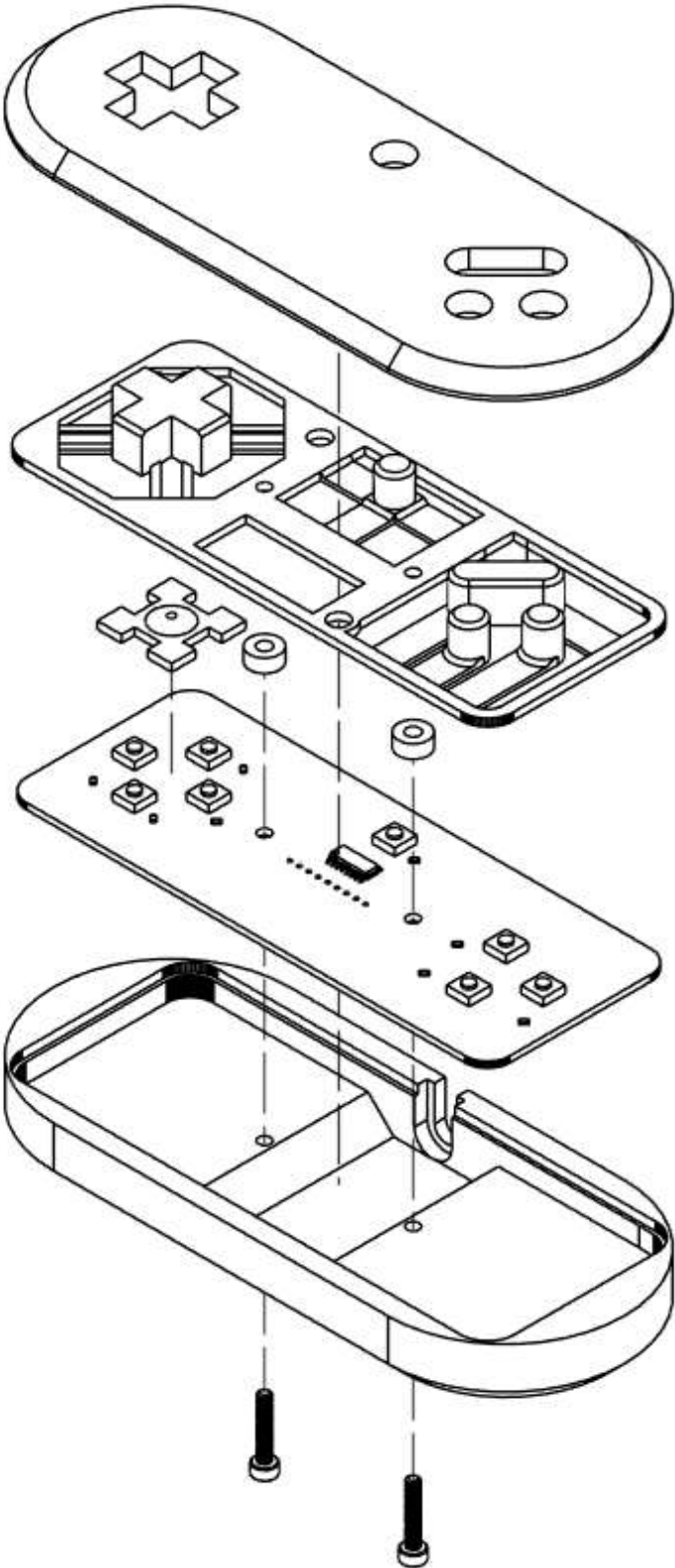
Figure 24: Fasten enclosure together with M3x20mm screws

Controller Wiring

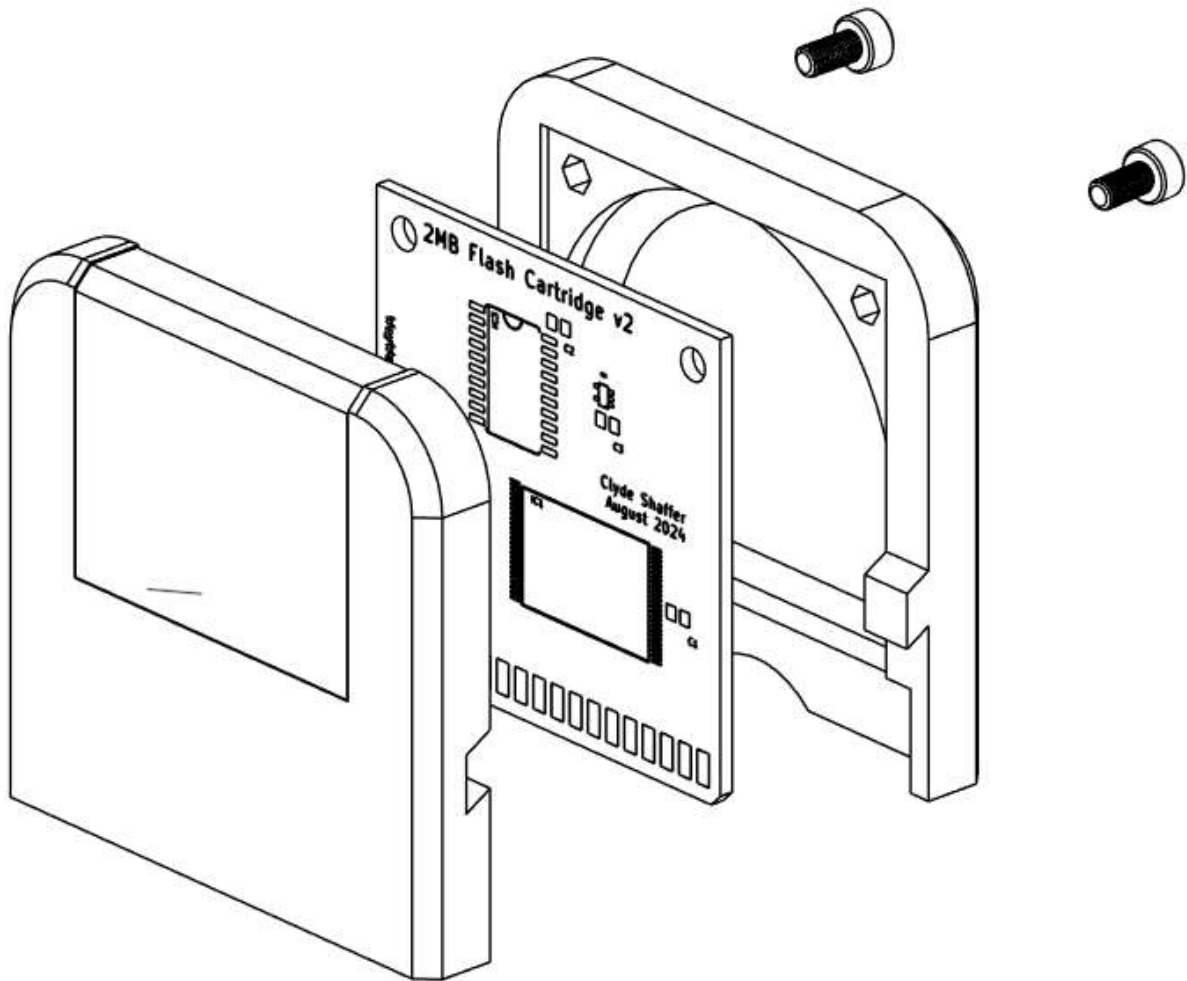


9 Wires soldered from cable, note the strain relief boot that seats into controller body

Controller Assembly

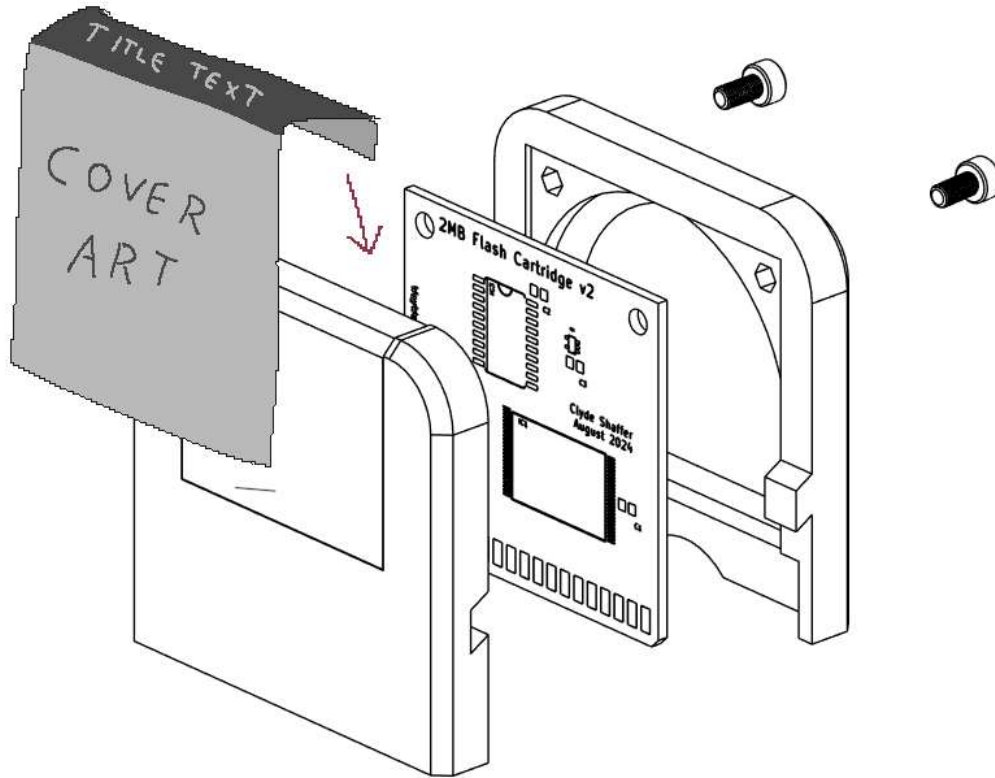


Data Cartridge



Two screws hold together the cartridge shell at the top, angled tabs stabilize the middle.

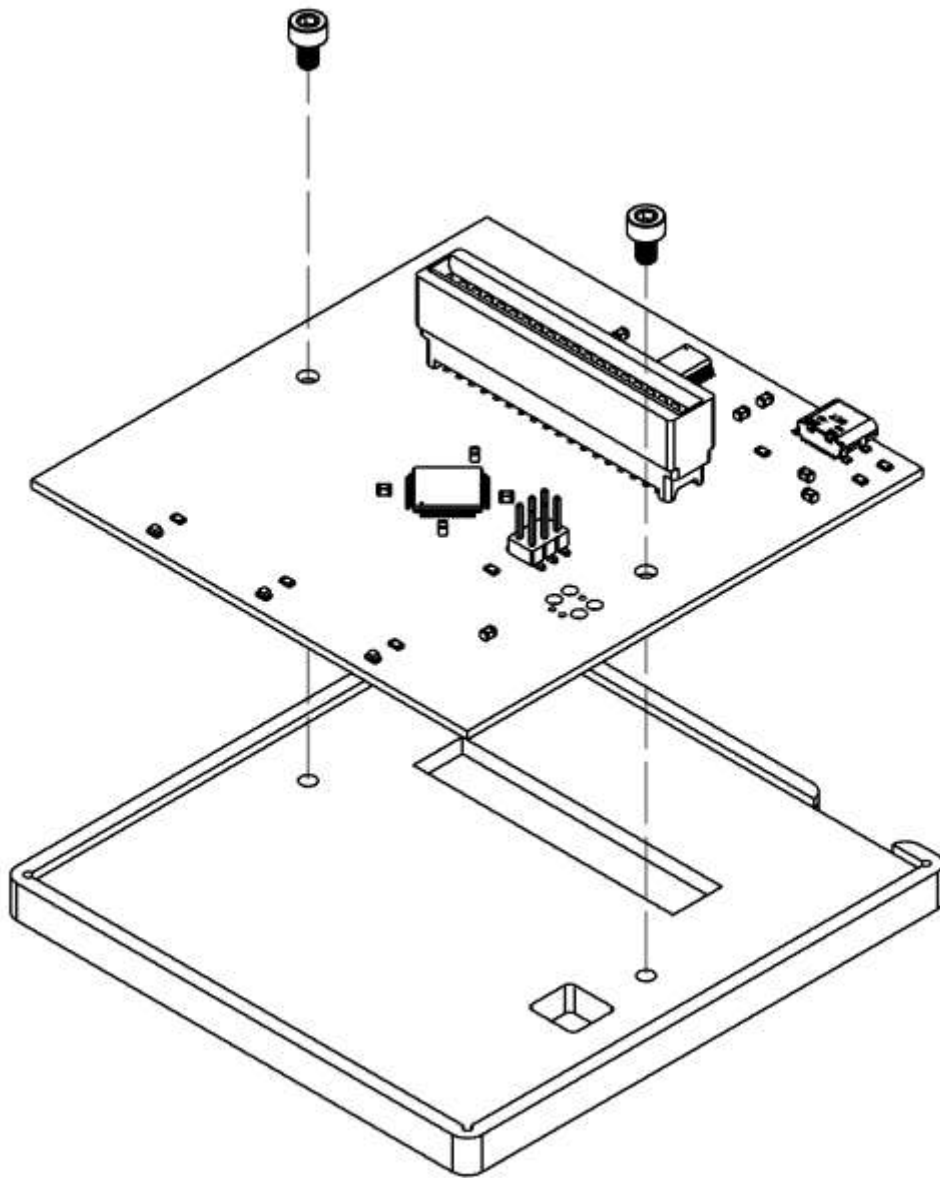
Data Cartridge, Labeled



Sticker for labeled cartridge aligns with scribe line on front of cartridge.

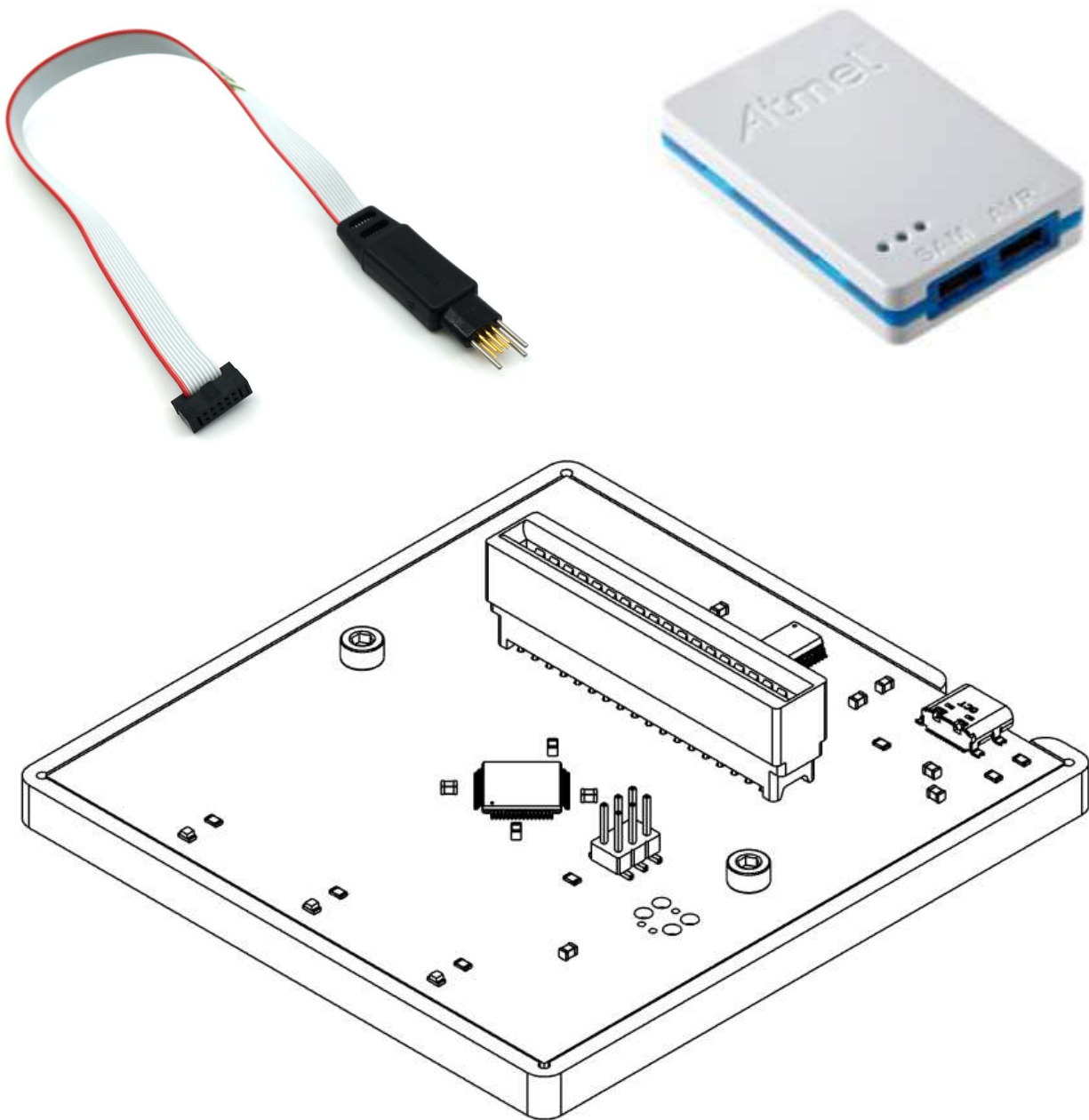
Cartridge label is printed slightly taller to fold over top and be clamped by cartridge body.

Cartridge Flasher Tool



Flasher board drops into base plate and fastens with M3x5 screws

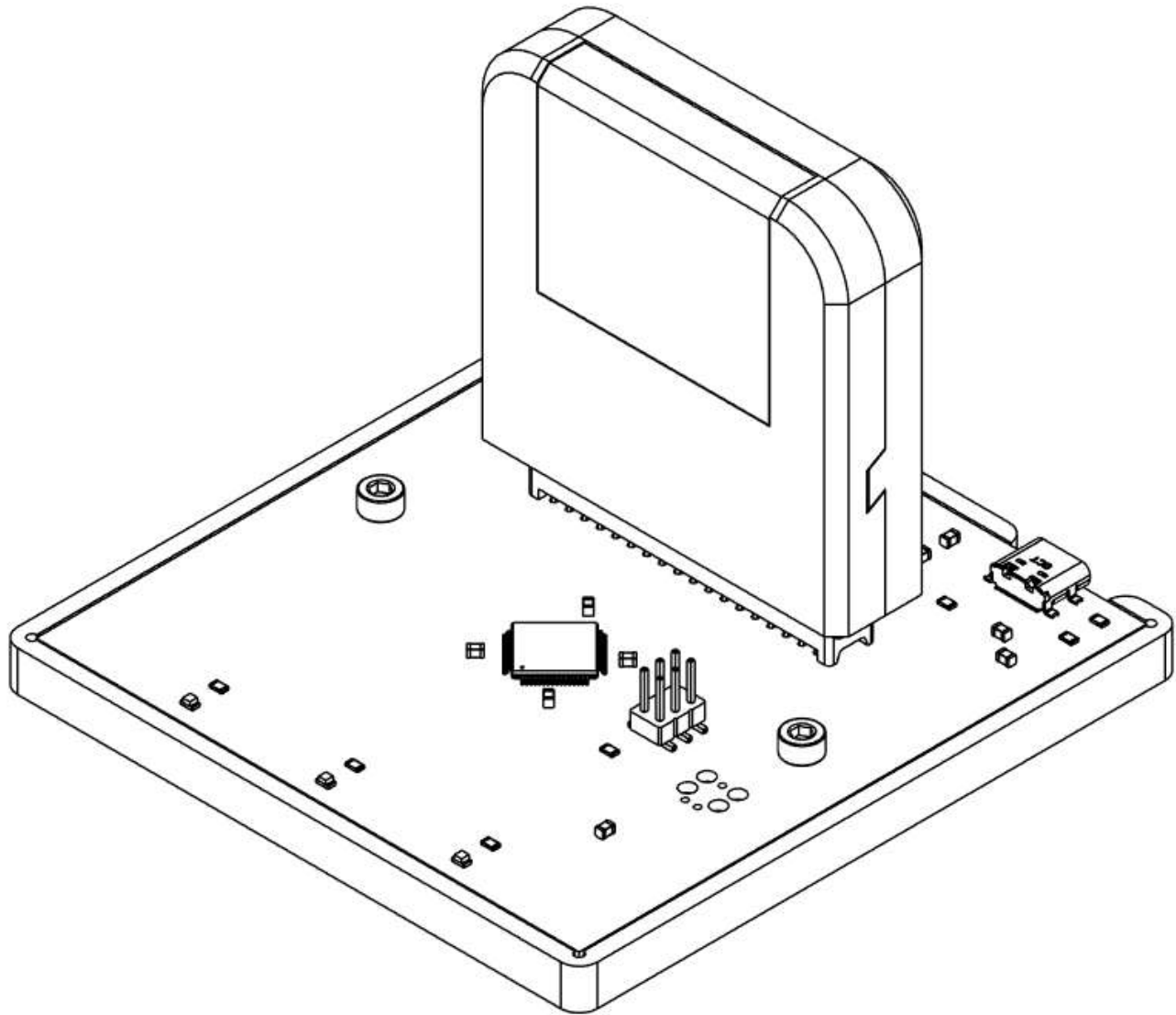
Cartridge Flasher Firmware



Tag Connect or generic header can be used for programming with Atmel-ICE. Optiboot firmware is flashed, then over USB-C avrdude is used to load the firmware. Afterward the cart flasher should respond to “help” command over serial.

Cartridge Flasher Operation

For cartridges to be labeled and shipped with data on them, the Cartridge Flasher should be used as follows:



- Cartridge is placed on edge connector, front side towards status LEDs
- Flasher connects to PC with USB-C cable and enumerates as a serial device (COM port on Windows)
- PC program “gtld” is used to initiate loading of the cartridge data

Parts Sourcing Notes

Dual-Ported SRAM CY7C1342-35JCT

Clydeware LLC will provide these.

IQD 28.6363MHz Oscillator

Clydeware LLC will provide these.

Controller Cable

Alibaba link: https://www.alibaba.com/product-detail/New-1-8M-6FT-9-Pin_1600240153607.html

A/V Cable

Alibaba link: https://www.alibaba.com/product-detail/RCA-TV-Composite-Video-Lead-1_1600507184687.html